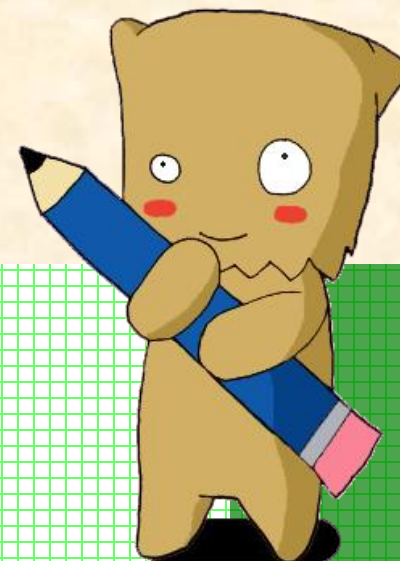




Chapter 8

STATISTICAL QUALITY CONTROL



Tanasanee Phienthrakul

Quality Improvement and Statistics

- Definitions of **Quality**
 - **Quality** means fitness for use
 - quality of design
 - quality of conformance
 - **Quality** is inversely proportional to variability.

Quality Improvement and Statistics

- Quality Improvement
 - **Quality improvement** is the reduction of variability in processes and products.
 - Alternatively, **quality improvement** is also seen as “waste reduction”.

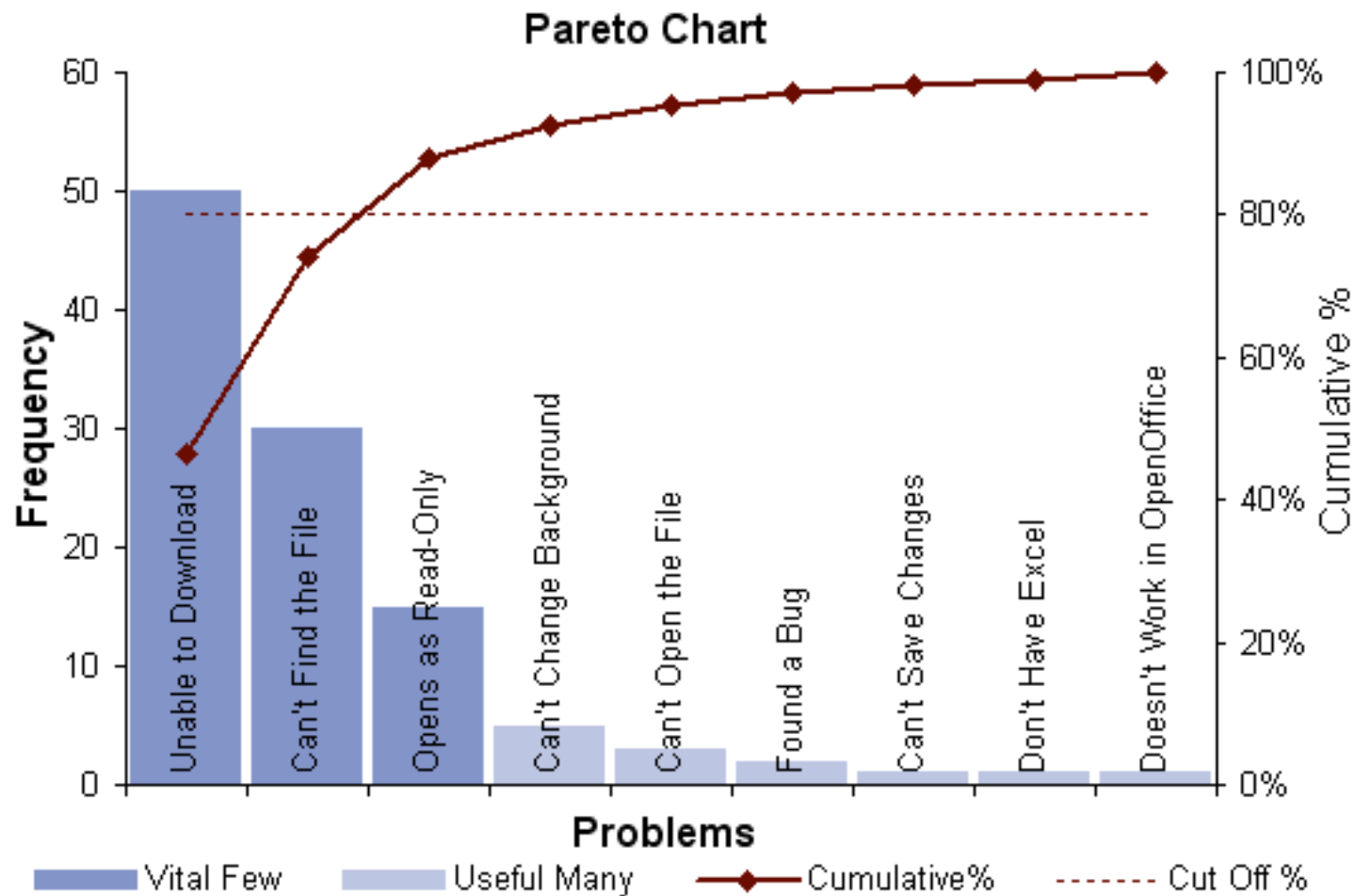
Statistical Process Control

- **Statistical process control** is a collection of tools that when used together can result in process stability and variance reduction

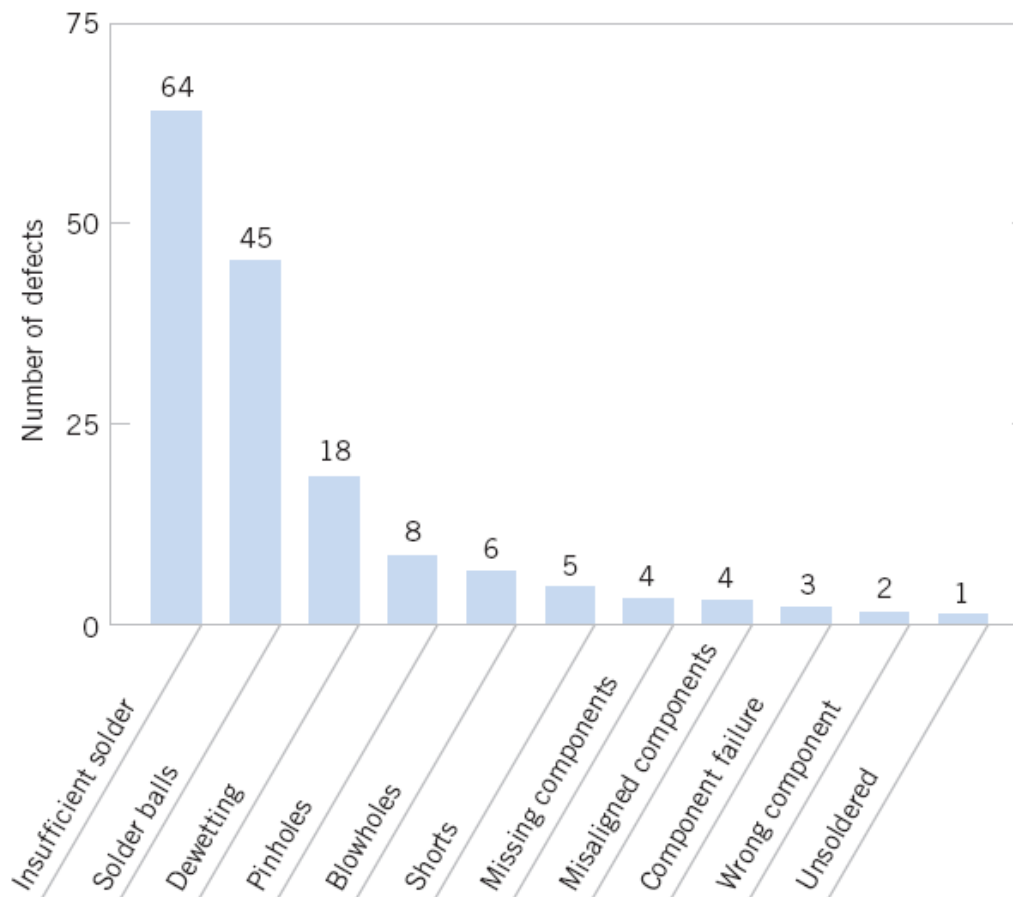
Statistical Process Control

- The **seven major tools** are
 - 1) Histogram
 - 2) Pareto Chart
 - 3) Cause and Effect Diagram
 - 4) Defect Concentration Diagram
 - 5) Control Chart
 - 6) Scatter Diagram
 - 7) Check Sheet

Histogram & Pareto Chart

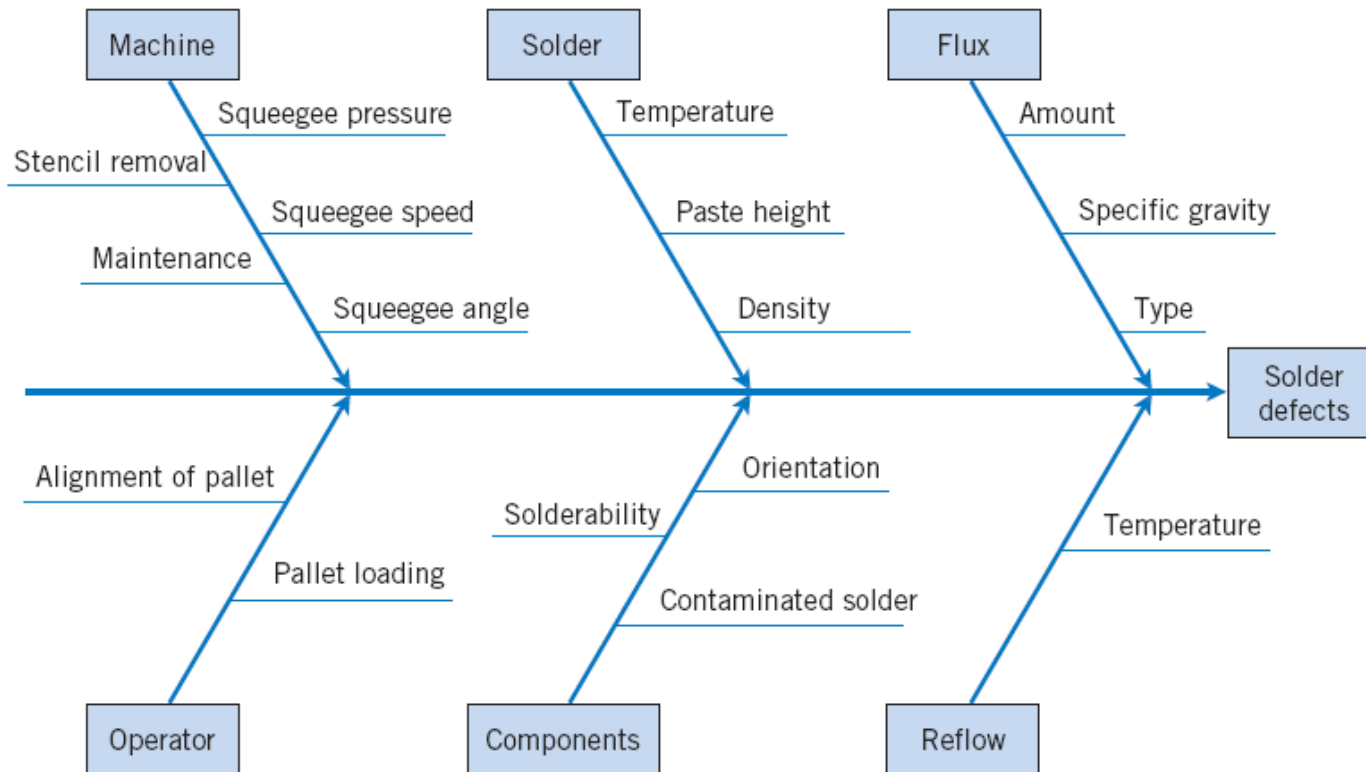


Pareto Diagram



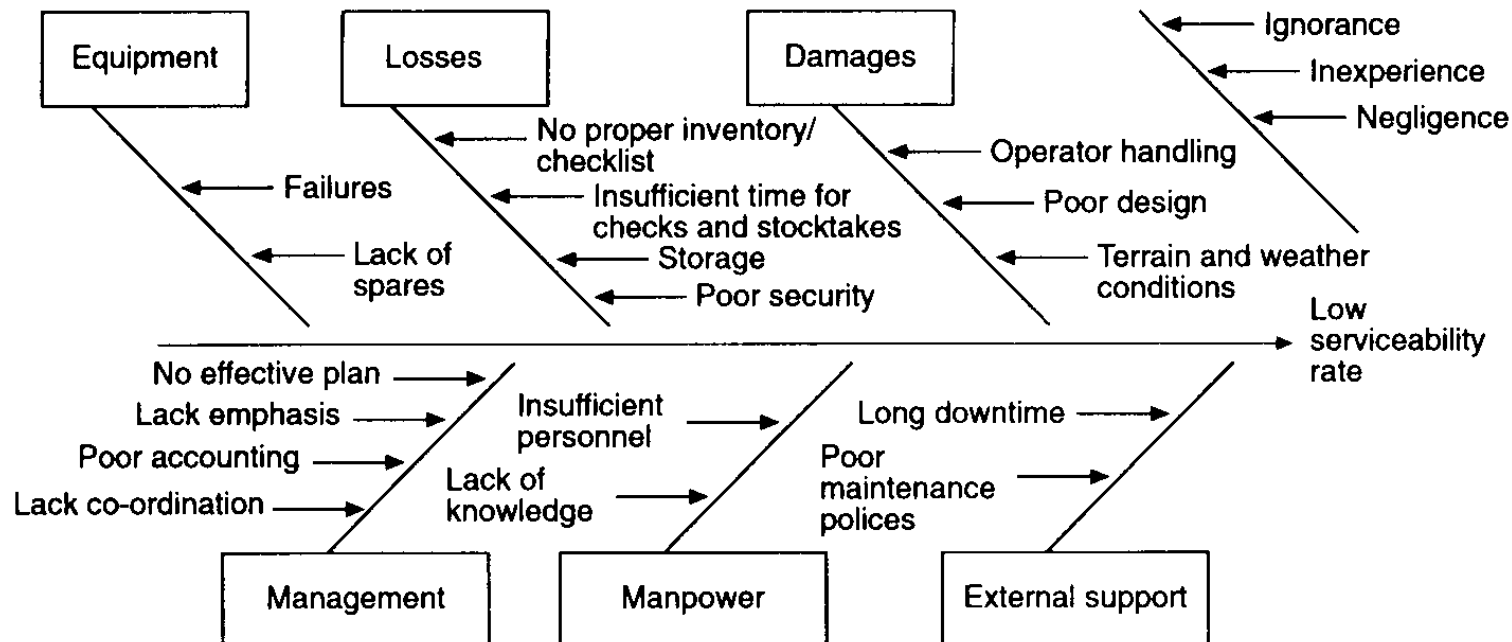
Pareto diagram for printed circuit board defects.

Cause-and-effect Diagram



Cause-and-effect diagram for the printed circuit board flow solder process.

Cause and Effect Diagram



Defect Concentration Diagram

Shaft Defect Concentration Diagram

Shaft No: 8567
 Lot Size: 1250
 Operator: KMB

Date: 4/23/91
 Total Defective: 26



Key for Defect Type

- Rust
- △ Burr
- * Scratch
- Pit

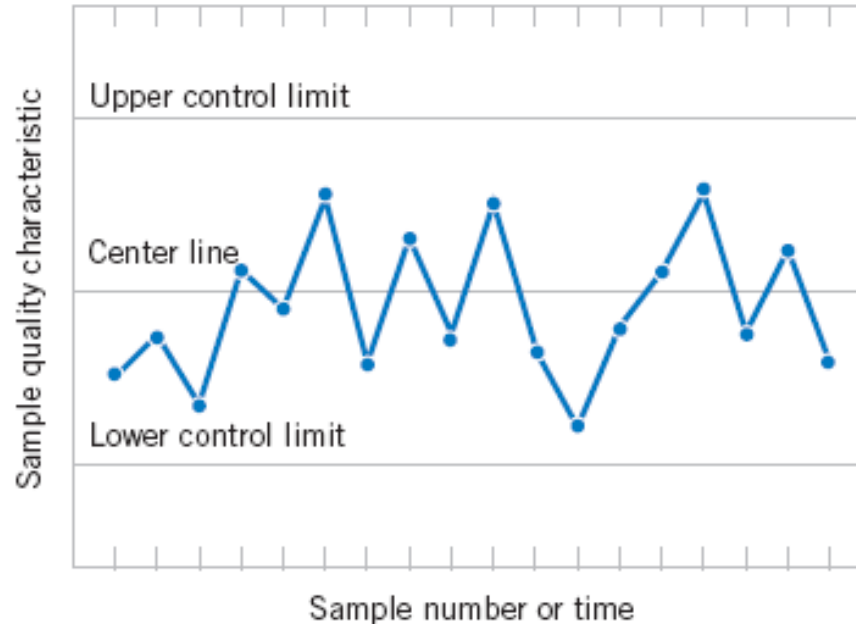
Introduction to Control Charts

Basic Principles

- A process that is operating with only *chance causes of variation* present is said to be **in statistical control**.
- A process that is operating in the presence of *assignable causes* is said to be **out of control**.
- The eventual goal of SPC is the *elimination of variability* in the process.

Introduction to Control Charts

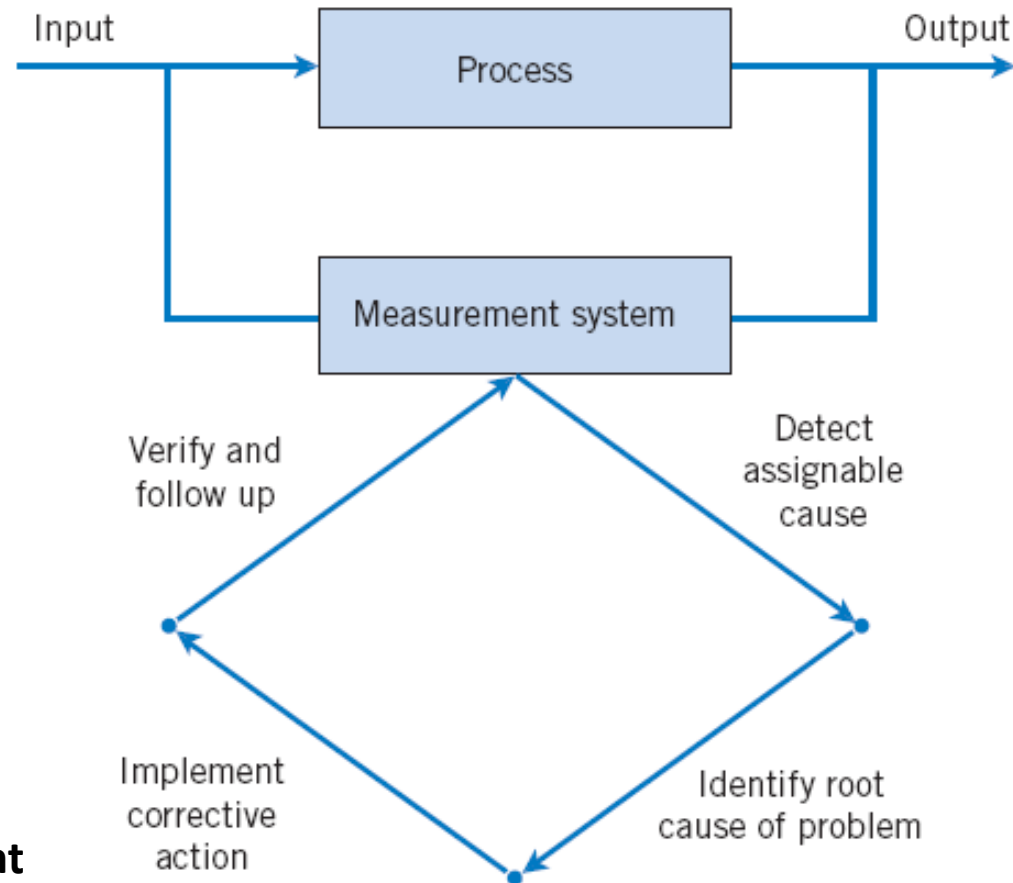
- Basic Principles
- A typical control chart has control limits set at values such that if the process is in control, nearly all points will lie within the upper control limit (UCL) and the lower control limit (LCL).



A typical control chart.

Introduction to Control Charts

- Basic Principles



**Process improvement
using the control chart.**

Introduction to Control Charts

- Basic Principles

$$UCL = \mu_W + k\sigma_W$$

$$CL = \mu_W$$

$$LCL = \mu_W - k\sigma_W$$

where

k = distance of the control limit from the center line

μ_W = mean of some sample statistic, W .

σ_W = standard deviation of some statistic, W .

Introduction to Control Charts

- Basic Principles

Important uses of the control chart

1. Most processes do not operate in a state of statistical control
2. Consequently, the routine and attentive use of control charts will identify assignable causes. If these causes can be eliminated from the process, variability will be reduced and the process will be improved
3. The control chart only **detects** assignable causes.
Management, operator, and engineering action will be necessary to eliminate the assignable causes.

Introduction to Control Charts

- Basic Principles

Types of control charts

- Variables Control Charts

- These charts are applied to data that follow a continuous distribution.

- Attributes Control Charts

- These charts are applied to data that follow a discrete distribution.

Introduction to Control Charts

- Basic Principles

Popularity of control charts

- 1) Control charts are a proven technique for improving productivity.
- 2) Control charts are effective in defect prevention.
- 3) Control charts prevent unnecessary process adjustment.
- 4) Control charts provide diagnostic information.
- 5) Control charts provide information about process capability.

Introduction to Control Charts

- Design of a Control Chart
- Suppose we have a process that we assume the true process mean is $\mu = 74$ and the process standard deviation is $\sigma = 0.01$. Samples of size 5 are taken giving a standard deviation of the sample average, is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.01}{\sqrt{5}} = 0.0045$$

Introduction to Control Charts

- Design of a Control Chart
- Control limits can be set at 3 standard deviations from the mean in both directions.
- “3–Sigma Control Limits”

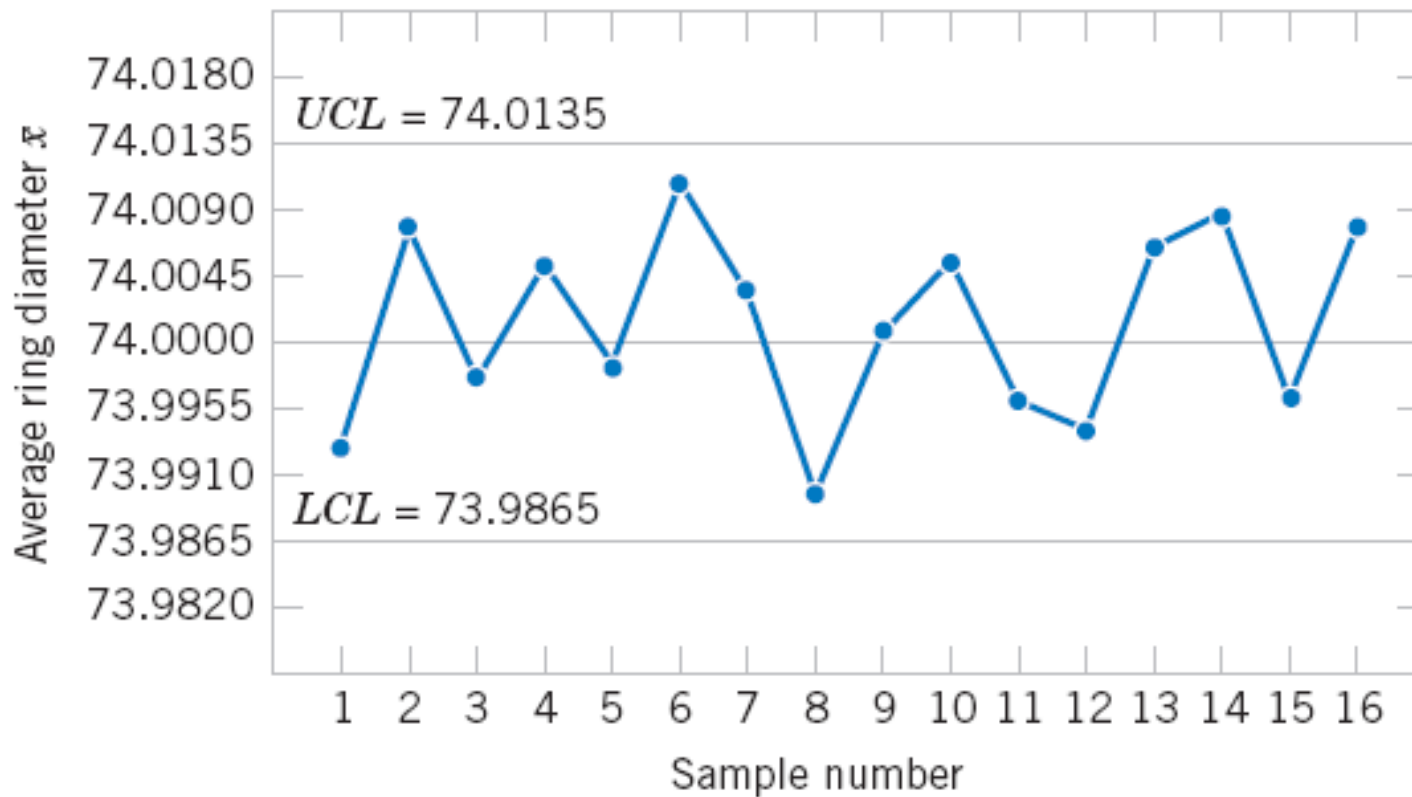
$$UCL = 74 + 3(0.0045) =$$

$$CL = 74$$

$$LCL = 74 - 3(0.0045) =$$

Introduction to Control Charts

- Design of a Control Chart



X-bar control chart for piston ring diameter.

Viscosity

- The viscosity of a chemical intermediate is measured every hour. Twenty samples consisting of a single observation are as follows.

Sample	Viscosity	Sample	Viscosity
1	378	11	462
2	438	12	502
3	487	13	449
4	515	14	470
5	485	15	501
6	474	16	470
7	486	17	512
8	548	18	530
9	502	19	462
10	440	20	491

- Find **3-Sigma Control Limits**

Introduction to Control Charts

- Design of a Control Chart
- Choosing the control limits is equivalent to setting up the critical region for hypothesis testing

$$H_0: \mu = 74$$

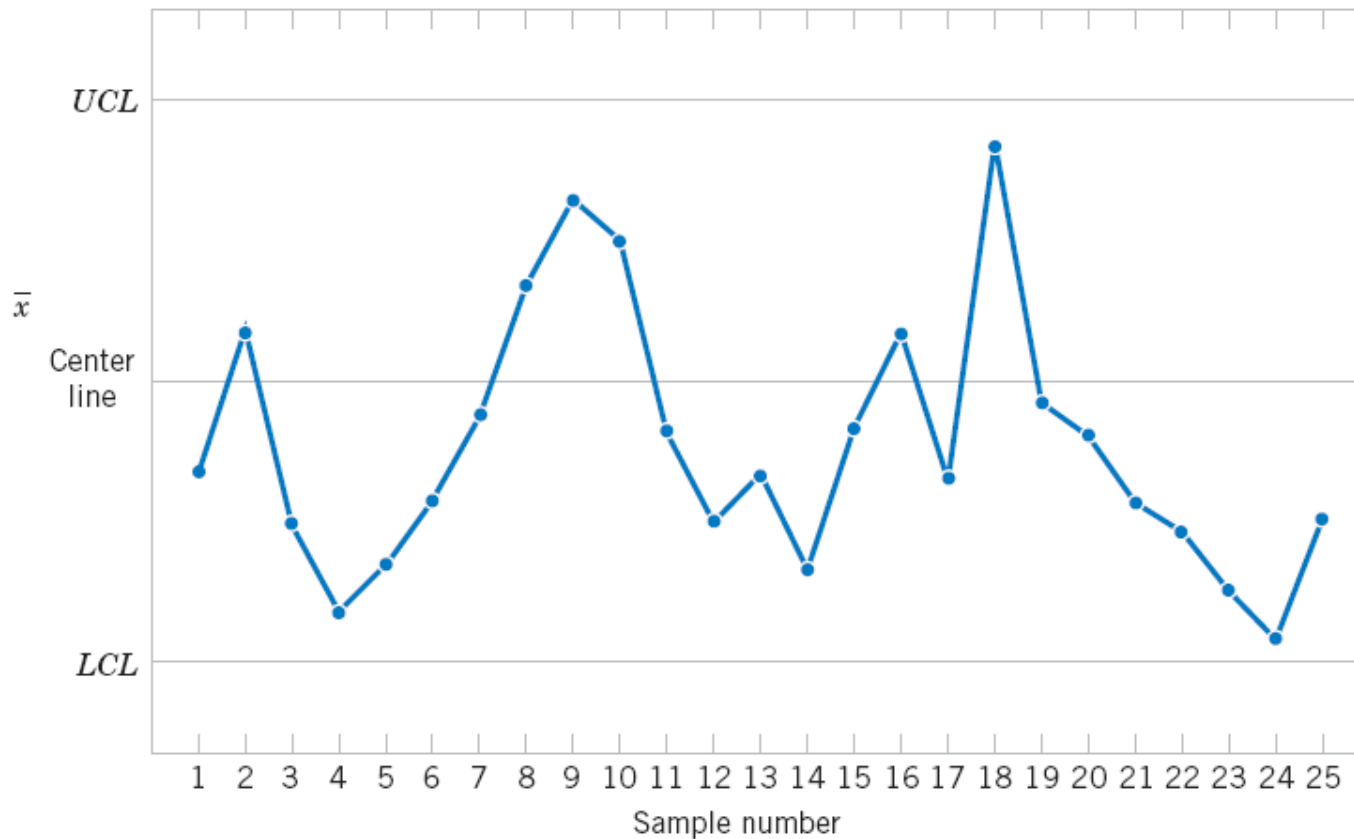
$$H_1: \mu \neq 74$$

Introduction to Control Charts

- Analysis of Patterns on Control Charts
- Look for “runs” – this is a sequence of observations of the same type (all above the center line, or all below the center line)
- Runs of say 8 observations or more could indicate an out-of-control situation.
 - Run up: a series of observations are increasing
 - Run down: a series of observations are decreasing

Introduction to Control Charts

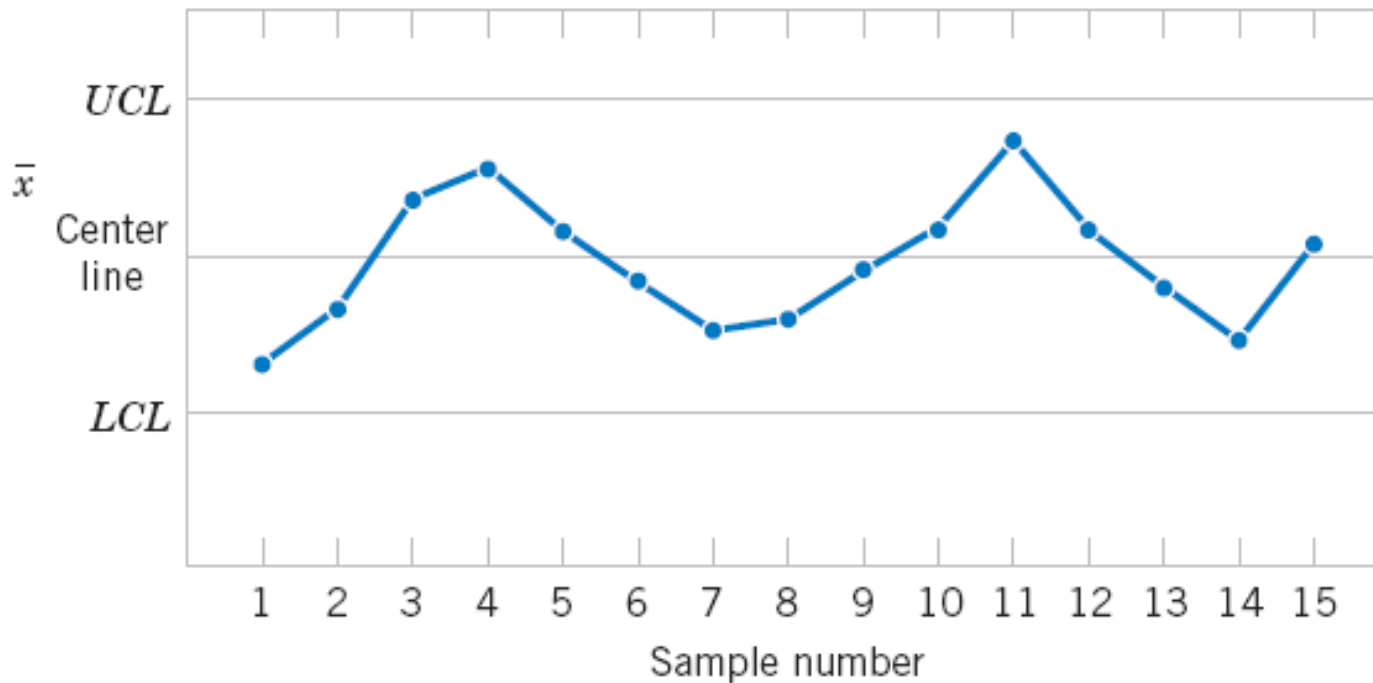
- Analysis of Patterns on Control Charts



***X-bar* control chart.**

Introduction to Control Charts

- Analysis of Patterns on Control Charts



An *X-bar* chart with a cyclic pattern

Introduction to Control Charts

- Analysis of Patterns on Control Charts

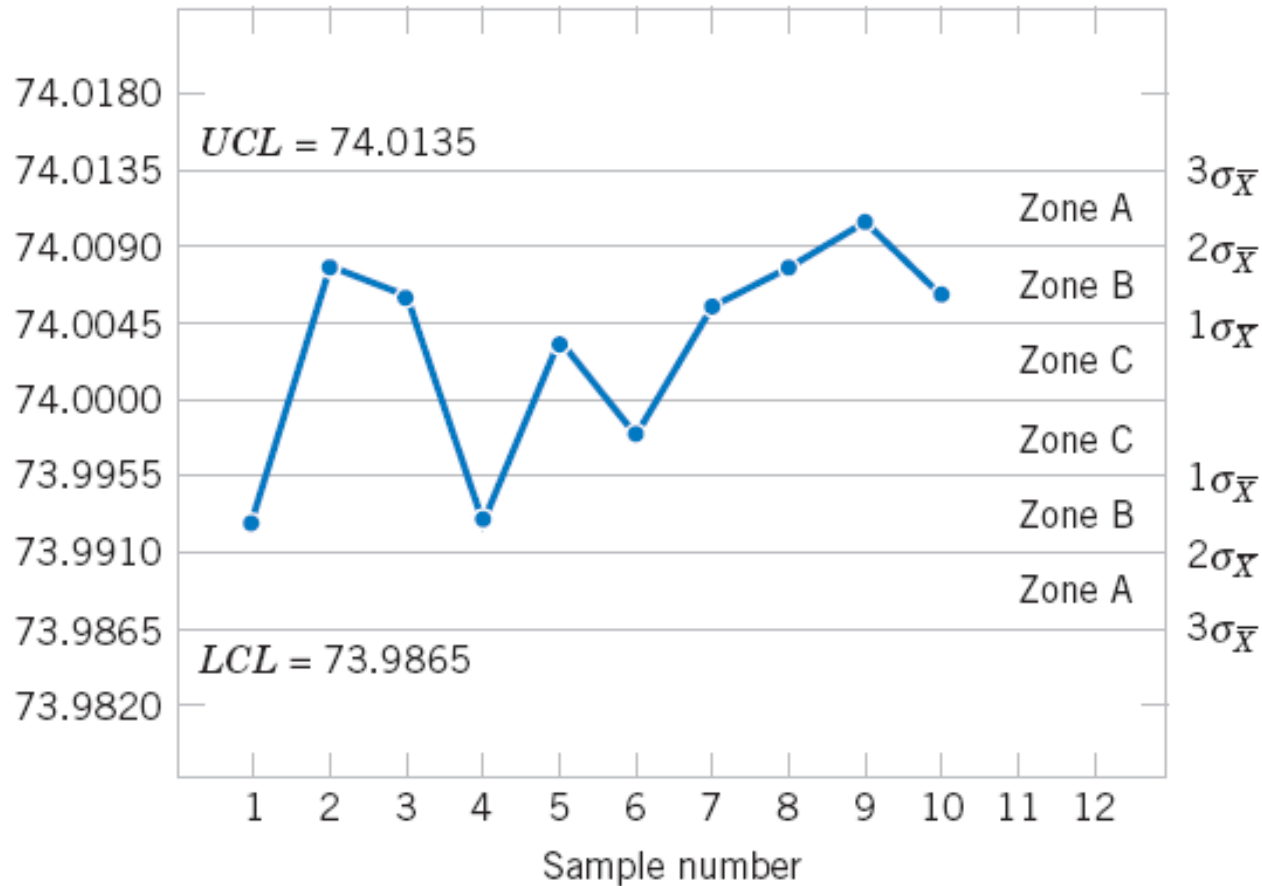
Western Electric Handbook Rules

A process is considered out of control if any of the following occur:

- 1) One point plots outside the 3-sigma control limits.
- 2) Two out of three consecutive points plot beyond the 2-sigma warning limits.
- 3) Four out of five consecutive points plot at a distance of 1-sigma or beyond from the center line.
- 4) Eight consecutive points plot on one side of the center line.

Introduction to Control Charts

- Analysis of Patterns on Control Charts



The Western Electric zone rules

X-bar and R or S Control Charts

- 3-sigma control limits:

$$UCL = \mu + 3\sigma/\sqrt{n}$$

$$LCL = \mu - 3\sigma/\sqrt{n}$$

$$CL = \mu$$

- The grand mean:

$$\hat{\mu} = \bar{\bar{X}} = \frac{1}{m} \sum_{i=1}^m \bar{X}_i$$

X-bar and R or S Control Charts

- The average range:

$$\bar{R} = \frac{1}{m} \sum_{i=1}^m R_i$$

- An unbiased estimator of σ :

$$\hat{\sigma} = \frac{\bar{R}}{d_2}$$

where the constant d_2 is tabulated for various sample sizes in Appendix Table X.

X-bar and R or S Control Charts

- $\bar{X}$ Control Chart (from $\bar{R}$):

The center line and upper and lower control limits for an $\bar{X}$ control chart are

$$UCL = \bar{\bar{x}} + A_2\bar{r} \quad CL = \bar{\bar{x}} \quad LCL = \bar{\bar{x}} - A_2\bar{r}$$

where the constant A_2 is tabulated for various sample sizes in Appendix Table X.

- R Chart:

The center line and upper and lower control limits for an R chart are

$$UCL = D_4\bar{r} \quad CL = \bar{r} \quad LCL = D_3\bar{r}$$

where $\bar{r}$ is the sample average range, and the constants D_3 and D_4 are tabulated for various sample sizes in Appendix Table X.

X-bar and R or S Control Charts

- 3-sigma control limits for S :

$$LCL = c_4\sigma - 3\sigma\sqrt{1 - c_4^2} \quad CL = c_4\sigma. \quad UCL = c_4\sigma + 3\sigma\sqrt{1 - c_4^2}$$

$$\bar{S} = \frac{1}{m} \sum_{i=1}^m S_i$$

- An unbiased estimator of σ :

$$\hat{\sigma} = \bar{S}/c_4$$

where the constant c_4 is tabulated for various sample sizes in Appendix Table X.

X-bar and R or S Control Charts

- S Chart:

$$UCL = \bar{s} + 3 \frac{\bar{s}}{c_4} \sqrt{1 - c_4^2} \quad CL = \bar{s} \quad LCL = \bar{s} - 3 \frac{\bar{s}}{c_4} \sqrt{1 - c_4^2}$$

- $\bar{X}$ Control Chart (from $\bar{S}$):

$$UCL = \bar{\bar{x}} + 3 \frac{\bar{s}}{c_4 \sqrt{n}} \quad CL = \bar{\bar{x}} \quad LCL = \bar{\bar{x}} - 3 \frac{\bar{s}}{c_4 \sqrt{n}}$$

X-bar and R or S Control Charts

A component part for a jet aircraft engine is manufactured by an investment casting process. The vane opening on this casting is an important functional parameter of the part. We will illustrate the use of $\bar{X}$ and R control charts to assess the statistical stability of this process.

Table presents 20 samples of five parts each. The values given in the table have been coded by using the last three digits of the dimension; that is, 31.6 should be 0.50316 inch. The quantities $\bar{\bar{x}} = 33.3$ and $\bar{\bar{r}} = 5.8$ are shown at the foot of Table .

Table Vane-Opening Measurements

Sample Number	x_1	x_2	x_3	x_4	x_5	$\bar{x}$	r	s
1	33	29	31	32	33	31.6	4	1.67332
2	33	31	35	37	31	33.4	6	2.60768
3	35	37	33	34	36	35.0	4	1.58114
4	30	31	33	34	33	32.2	4	1.64317
5	33	34	35	33	34	33.8	2	0.83666
6	38	37	39	40	38	38.4	3	1.14018
7	30	31	32	34	31	31.6	4	1.51658
8	29	39	38	39	39	36.8	10	4.38178
9	28	33	35	36	43	35.0	15	5.43139
10	38	33	32	35	32	34.0	6	2.54951
11	28	30	28	32	31	29.8	4	1.78885
12	31	35	35	35	34	34.0	4	1.73205
13	27	32	34	35	37	33.0	10	3.80789
14	33	33	35	37	36	34.8	4	1.78885
15	35	37	32	35	39	35.6	7	2.60768
16	33	33	27	31	30	30.8	6	2.48998
17	35	34	34	30	32	33.0	5	2.00000
18	32	33	30	30	33	31.6	3	1.51658
19	25	27	34	27	28	28.2	9	3.42053
20	35	35	36	33	30	33.8	6	2.38747
						$\bar{\bar{x}} = 33.32$	$\bar{r} = 5.8$	$\bar{s} = 2.345$

X-bar and R or S Control Charts

The value of A_2 for samples of size 5 is $A_2 = 0.577$. Then the trial control limits for the $\bar{X}$ chart are

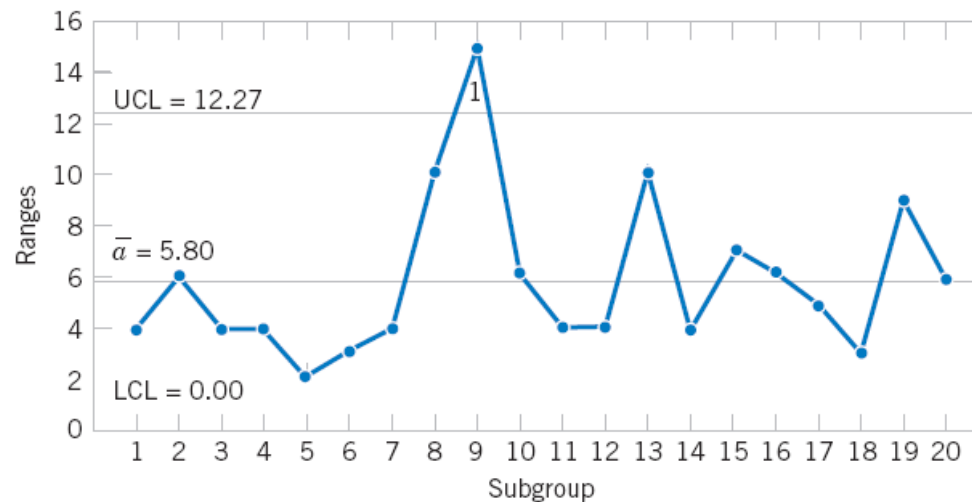
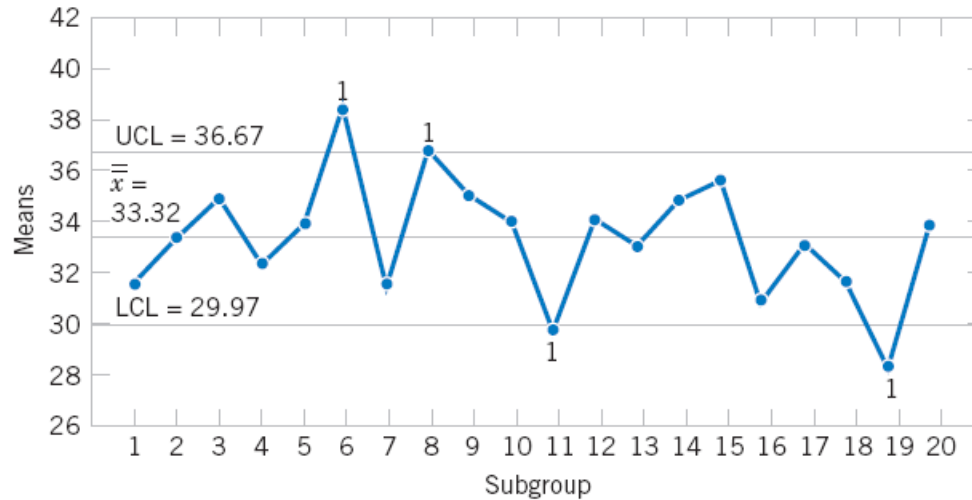
For the R chart, the trial control limits are

$$UCL = D_4 \bar{r} =$$

$$LCL = D_3 \bar{r} =$$

The $\bar{X}$ and R control charts with these trial control limits are shown in Fig. Notice that samples 6, 8, 11, and 19 are out of control on the $\bar{X}$ chart and that sample 9 is out of control on the R chart. (These points are labeled with a “1” because they violate the first Western Electric rule.)

X-bar and R or S Control Charts



$\bar{X}$ and R control charts for vane opening.

X-bar and R or S Control Charts

For the S chart, the value of $c_4 =$ Therefore,

$$\frac{3\bar{s}}{c_4} \sqrt{1 - c_4^2} =$$

and the trial control limits are

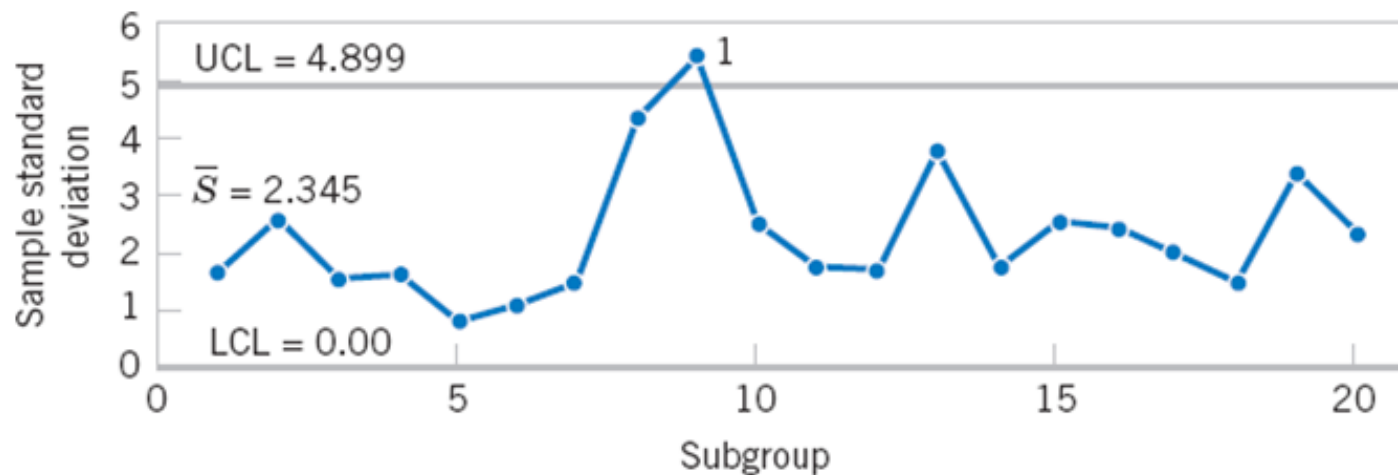
$$UCL =$$

$$LCL =$$

The LCL is set to zero. If $\bar{s}$ is used to determine the control limits for the $\bar{X}$ chart,

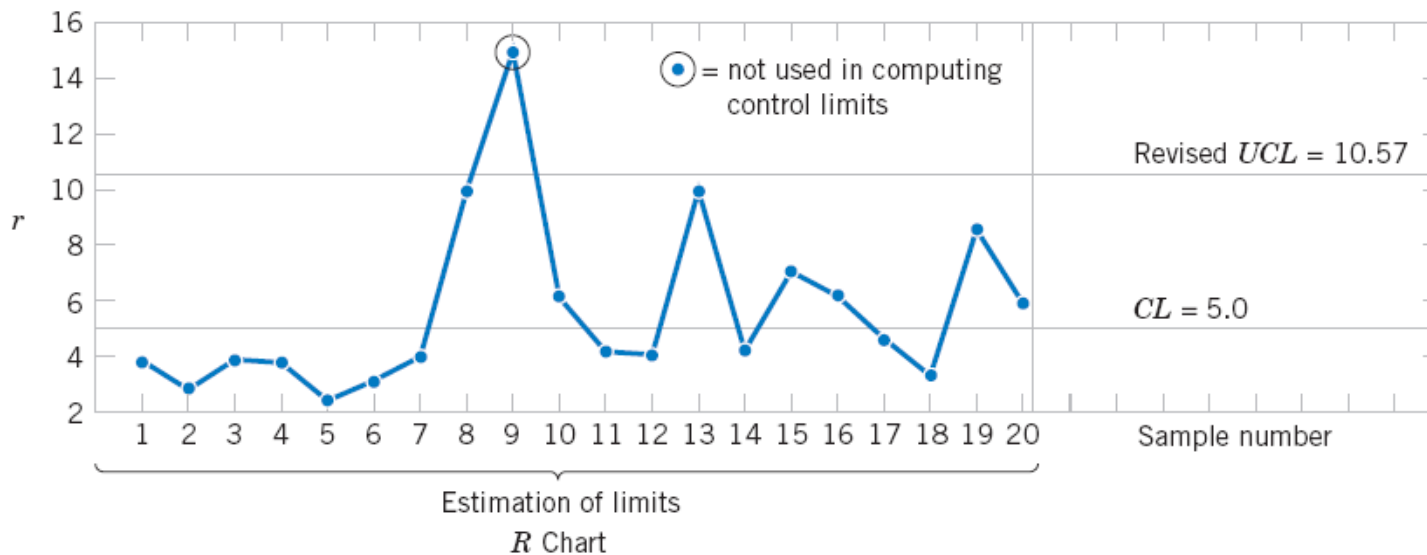
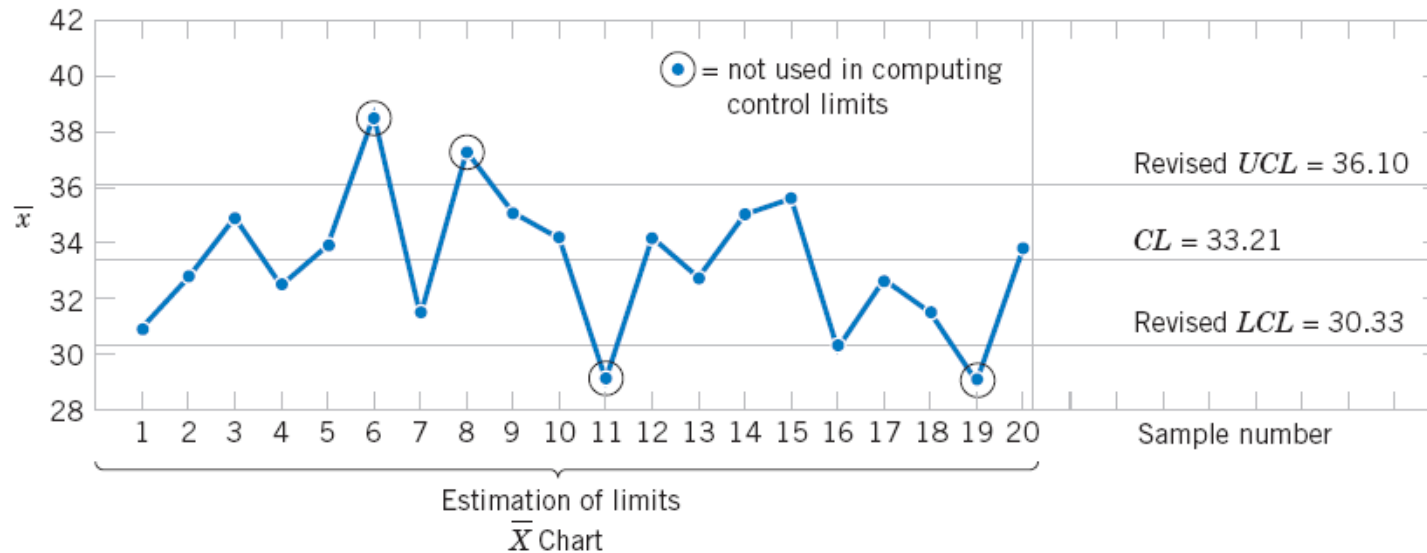
$$\bar{\bar{x}} \pm \frac{3\bar{s}}{c_4 \sqrt{n}} =$$

X-bar and R or S Control Charts



The S control chart for vane opening.

The $\bar{X}$ -bar and R control charts for vane opening.



Control Charts for Individual Measurements

- What if you could not get a sample size greater than 1 ($n = 1$)?
- Examples include
 - Automated inspection and measurement technology is used, and every unit manufactured is analyzed.
 - The production rate is very slow, and it is inconvenient to allow samples sizes of $N > 1$ to accumulate before analysis
 - Repeat measurements on the process differ only because of laboratory or analysis error, as in many chemical processes.
- The individual control charts are useful for samples of sizes $n = 1$.

Control Charts for Individual Measurements

- The *moving range* (MR) is defined as the absolute difference between two successive observations:

$$MR_i = |x_i - x_{i-1}|$$

which will indicate possible shifts or changes in the process from one observation to the next.

Control Charts for Individual Measurements

- Individuals Control Chart

The center line and upper and lower control limits for a control chart for individuals are

$$UCL = \bar{x} + 3 \frac{\overline{mr}}{d_2} = \bar{x} + 3 \frac{\overline{mr}}{1.128}$$

$$CL = \bar{x}$$

$$LCL = \bar{x} - 3 \frac{\overline{mr}}{d_2} = \bar{x} - 3 \frac{\overline{mr}}{1.128}$$

and for a control chart for moving ranges

$$UCL = D_4 \overline{mr} = 3.267 \overline{mr}$$

$$CL = \overline{mr}$$

$$LCL = D_3 \overline{mr} = 0$$

Concentration

- The concentration of a chemical process is measured every hour. Twenty samples consisting of a single observation are as follows.

Sample	Concentration	Sample	Concentration
1	102.0	11	101.3
2	94.8	12	98.7
3	98.3	13	101.1
4	98.4	14	98.4
5	102.0	15	97.0
6	98.5	16	96.7
7	99.0	17	100.3
8	97.7	18	101.4
9	100.0	19	97.2
10	98.1	20	101.0

- Find $\bar{X}$ and $\overline{MR}$ control charts

Control Charts for Individual Measurements

Interpretation of the Charts

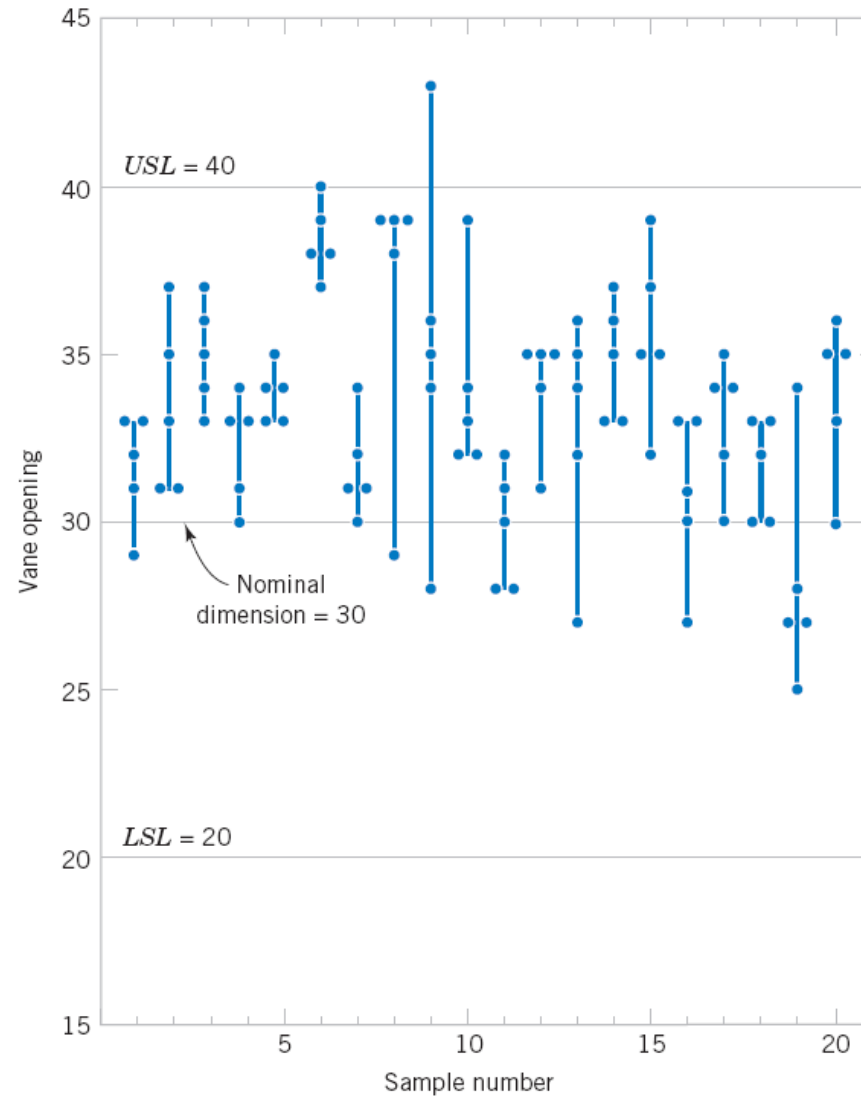
- X Charts can be interpreted similar to *X-bar* charts. MR charts cannot be interpreted the same as *X-bar* or R charts.
- Since the MR chart plots data that are “correlated” with one another, then **looking for patterns on the chart does not make sense.**
- MR chart **cannot really supply useful information about process variability.**
- More emphasis should be placed on interpretation of the X chart.

Process Capability

- **Process capability** refers to the performance of the process when it is operating in control.
- Two graphical tools are helpful in assessing process capability:
 - **Tolerance chart** (or tier chart)
 - **Histogram**

Process Capability

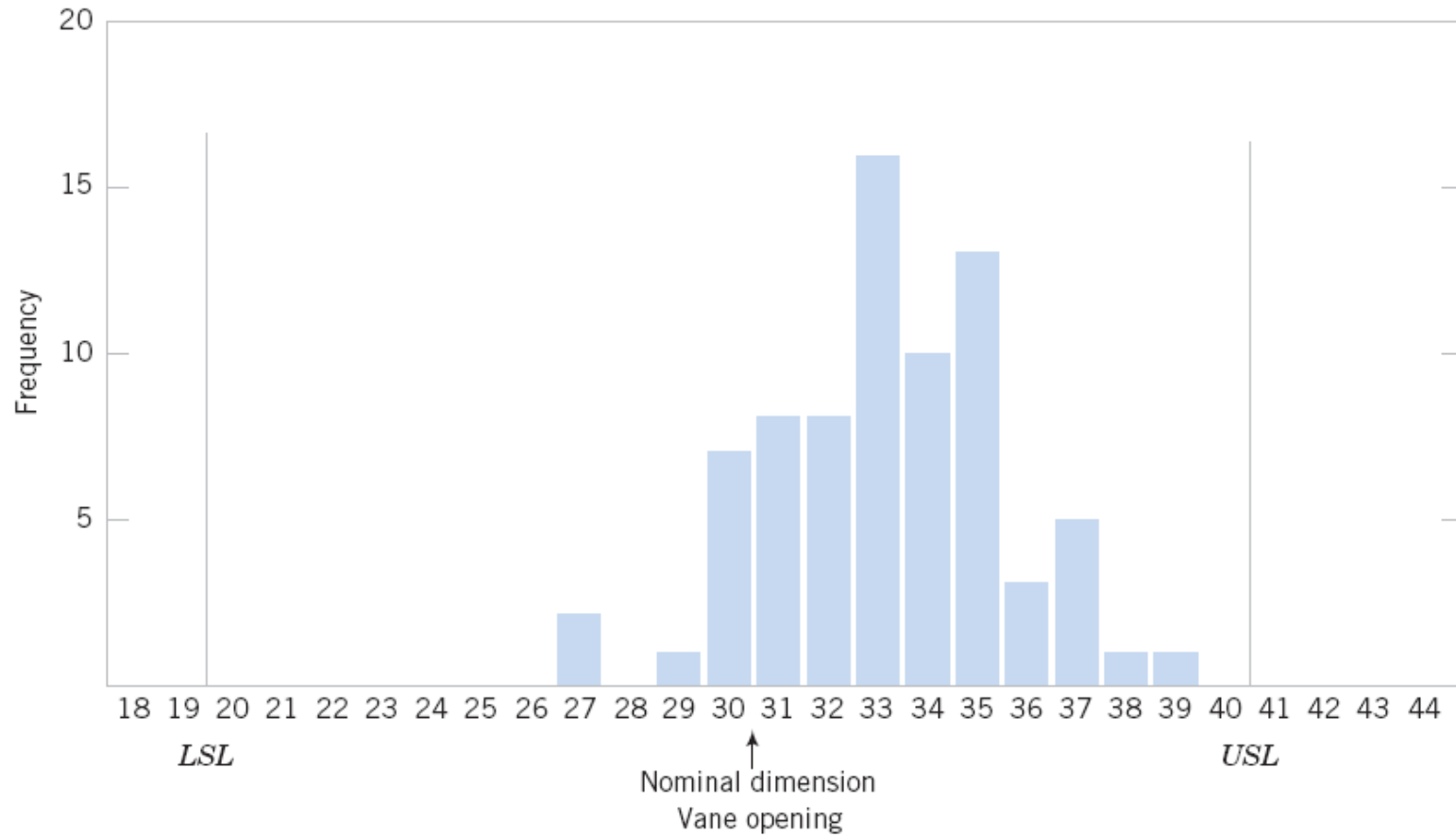
- Tolerance Chart



Tolerance diagram
of vane openings.

Process Capability

- Histogram



Histogram for vane openings.

Process Capability

- Process Capability Ratio

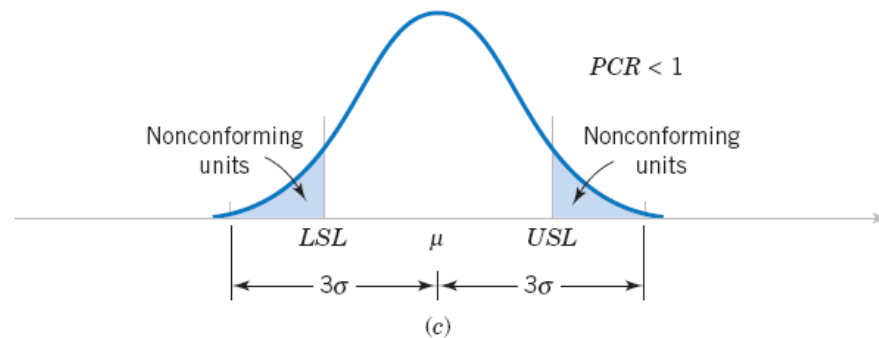
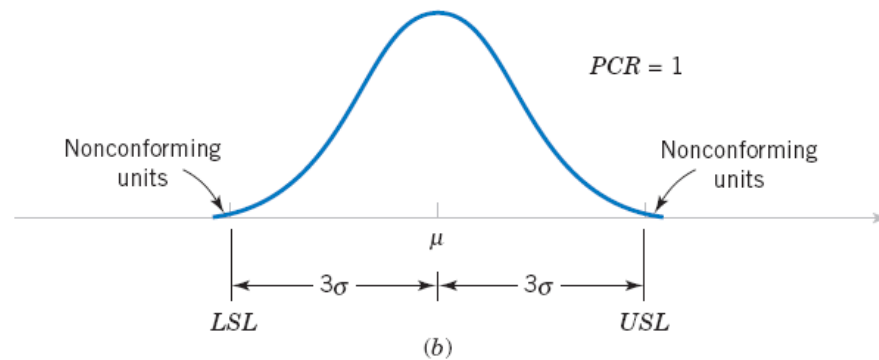
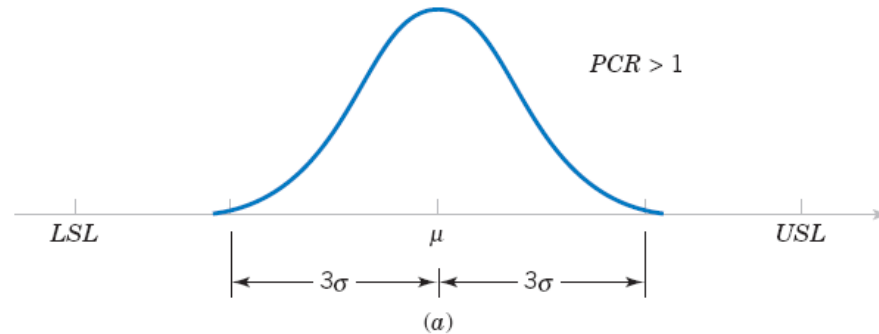
The **process capability ratio** (PCR) is

$$PCR = \frac{USL - LSL}{6\sigma}$$

- One-sided Process Capability Ratio – PCR_k

$$PCR_k = \min \left[\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma} \right]$$

Process Capability



Process Fallout and the process capability ratio (PCR).

Process Capability

Electrical Current

For an electronic manufacturing process a current has specifications of 100 ± 10 milliamperes. The process mean μ and standard deviation σ are 107.0 and 1.5, respectively. The process mean is nearer to the *USL*. Consequently,

$$PCR =$$

and

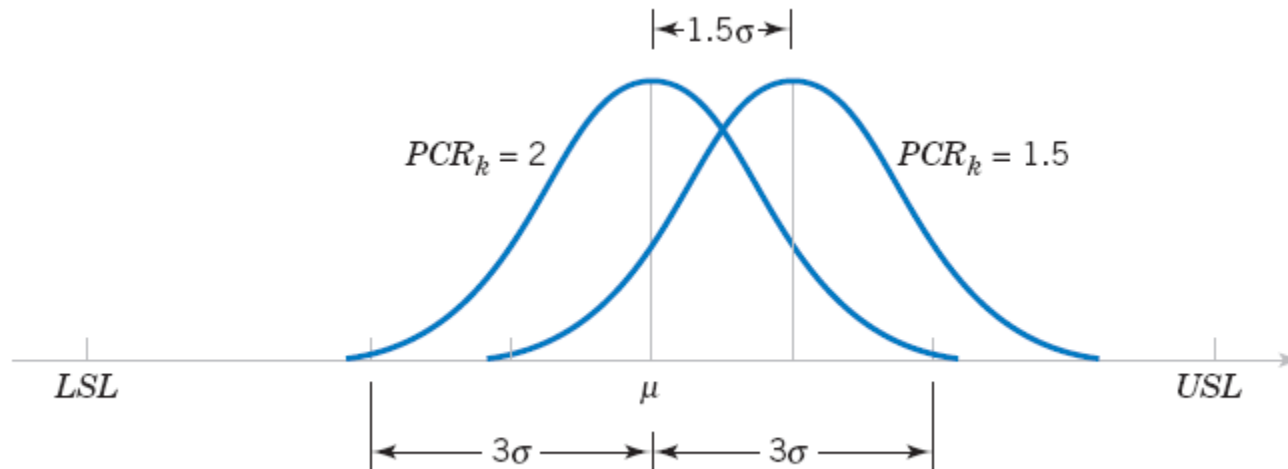
$$PCR_k =$$

The small PCR_k indicates that the process is likely to produce currents outside of the specification limits. From the normal distribution,

$$\begin{aligned} P(X < LSL) &= P(Z < (90 - 107)/1.5) \\ &= P(Z < -11.33) \cong \end{aligned}$$

$$\begin{aligned} P(X > USL) &= P(Z > (110 - 107)/1.5) \\ &= P(Z > 2) = \end{aligned}$$

Process Capability



Mean of a six-sigma process shifts by 1.5 standard deviations.

Attribute Control Charts

P Chart (Control Chart for Proportions)

The center line and upper and lower control limits for the *P* chart are

$$UCL = \bar{p} + 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}} \quad CL = \bar{p} \quad LCL = \bar{p} - 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

where $\bar{p}$ is the observed value of the average fraction defective.

Attribute Control Charts

Ceramic Substrate

Suppose we wish to construct a fraction-defective control chart for a ceramic substrate production line. We have 20 preliminary samples, each of size 100; the number of defectives in each sample is shown in Table. Assume that the samples are numbered in the sequence of production. Note that $\bar{p} = (800/2000) = 0.40$; therefore, the trial parameters for the control chart are

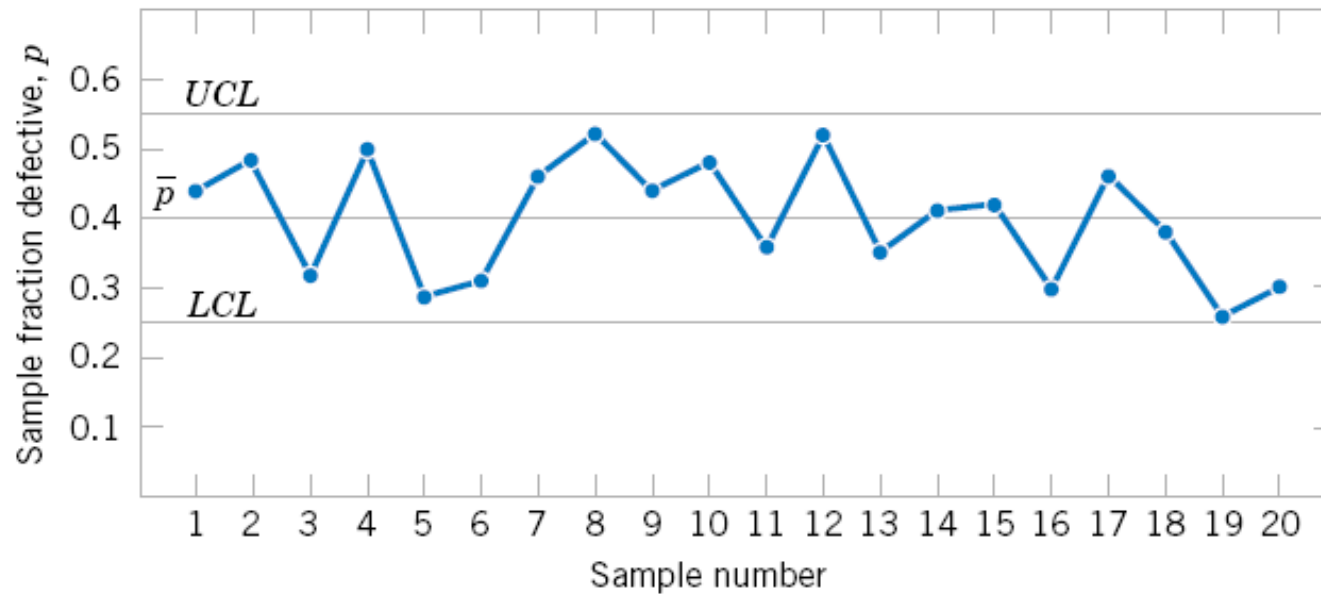
$$UCL =$$

$$CL =$$

$$LCL =$$

The control chart is shown in Fig. All samples are in control. If they were not, we would search for assignable causes of variation and revise the limits accordingly. This chart can be used for controlling future production.

Attribute Control Charts



P chart for a ceramic substrate.

Attribute Control Charts

Practical Interpretation: Although this process exhibits statistical control, its defective rate ($\bar{p} = 0.40$) is very poor. We should take appropriate steps to investigate the process to determine why such a large number of defective units is being produced. Defective units should be analyzed to determine the specific types of defects present. Once the defect types are known, process changes should be investigated to determine their impact on defect levels. Designed experiments may be useful in this regard.

Attribute Control Charts

U Chart (Control Chart for Defects per Unit)

The center line and upper and lower control limits on the U chart are

$$UCL = \bar{u} + 3\sqrt{\frac{\bar{u}}{n}} \quad CL = \bar{u} \quad LCL = \bar{u} - 3\sqrt{\frac{\bar{u}}{n}}$$

where $\bar{u}$ is the average number of defects per unit.

Attribute Control Charts

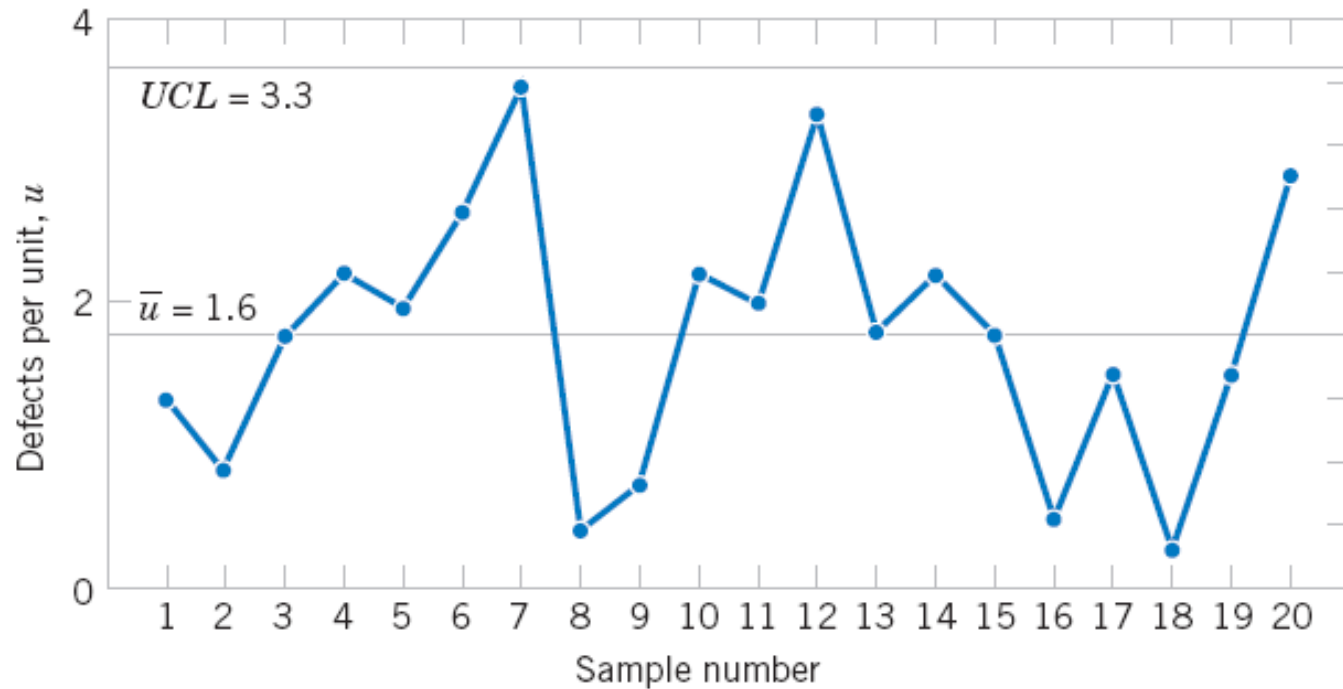
Printed Circuit Boards

Printed circuit boards are assembled by a combination of manual assembly and automation. Surface Mount Technology (SMT) is used to make the mechanical and electrical connections of the components to the board. Every hour, five boards are selected and inspected for process-control purposes. The number of defects in each sample of five boards is noted. Results for 20 samples are shown in Table

Number of Defects in Samples of Five Printed Circuit Boards

Sample	Number of Defects	Defects per Unit u_i	Sample	Number of Defects	Defects per Unit u_i
1	6	1.2	11	9	1.8
2	4	0.8	12	15	3.0
3	8	1.6	13	8	1.6
4	10	2.0	14	10	2.0
5	9	1.8	15	8	1.6
6	12	2.4	16	2	0.4
7	16	3.2	17	7	1.4
8	2	0.4	18	1	0.2
9	3	0.6	19	7	1.4
10	10	2.0	20	13	2.6

Attribute Control Charts



U chart of defects per unit on printed circuit boards.

Time-Weighted Charts

Cumulative Sum Control Chart

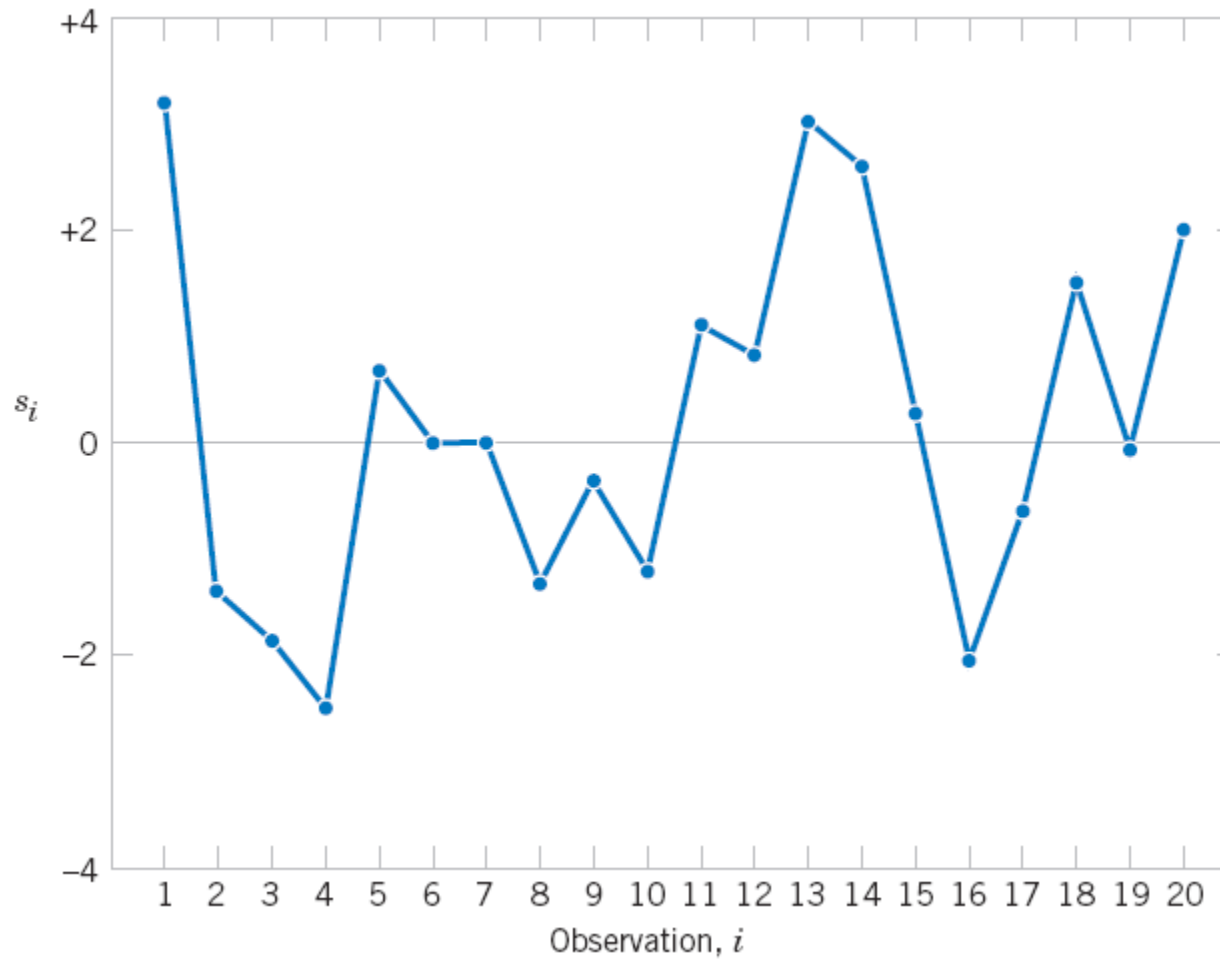
- The **cusum chart** incorporates all information in the sequence of sample values by plotting the **cumulative sums** of the deviations of the sample values from a target value.
- If μ_0 is the target for the process mean, $\bar{X}_j$ is the average of the j th sample, then the cumulative sum control chart is formed by plotting the quantity

$$S_i = \sum_{j=1}^i (\bar{X}_j - \mu_0)$$

CUSUM Computations for the Chemical Process Concentration Data

Observation, i	x_i	$x_i - 99$	$s_i = (x_i - 99) + s_{i-1}$
1	102.0	3.0	3.0
2	94.8	-4.2	-1.2
3	98.3	-0.7	-1.9
4	98.4	-0.6	-2.5
5	102.0	3.0	0.5
6	98.5	-0.5	0.0
7	99.0	0.0	0.0
8	97.7	-1.3	-1.3
9	100.0	1.0	-0.3
10	98.1	-0.9	-1.2
11	101.3	2.3	1.1
12	98.7	-0.3	0.8
13	101.1	2.1	2.9
14	98.4	-0.6	2.3
15	97.0	-2.0	0.3
16	96.7	-2.3	-2.0
17	100.3	1.3	-0.7
18	101.4	2.4	1.7
19	97.2	-1.8	-0.1
20	101.0	2.0	1.9

Time-Weighted Charts



Plot of the cumulative sum for the concentration data

Time-Weighted Charts

Exponential Weighted Moving Average Control Chart

Data collected in time order is often averaged over several time periods. For example, economic data is often presented as an average over the last four quarters. That is, at time t the average of the last four measurements can be written as

$$\bar{x}_t(4) = \frac{1}{4}x_t + \frac{1}{4}x_{t-1} + \frac{1}{4}x_{t-2} + \frac{1}{4}x_{t-3}$$

This average places weight of $1/4$ on each of the most recent observations, and zero weight on older observations. It is called a **moving average** and in this case a *window* of size 4 is used.

Concentration

- The concentration of a chemical process is measured every hour. Twenty samples consisting of a single observation are as follows.

Sample	Concentration	Sample	Concentration
1	102.0	11	101.3
2	94.8	12	98.7
3	98.3	13	101.1
4	98.4	14	98.4
5	102.0	15	97.0
6	98.5	16	96.7
7	99.0	17	100.3
8	97.7	18	101.4
9	100.0	19	97.2
10	98.1	20	101.0

Time-Weighted Charts

Exponential Weighted Moving Average Control Chart

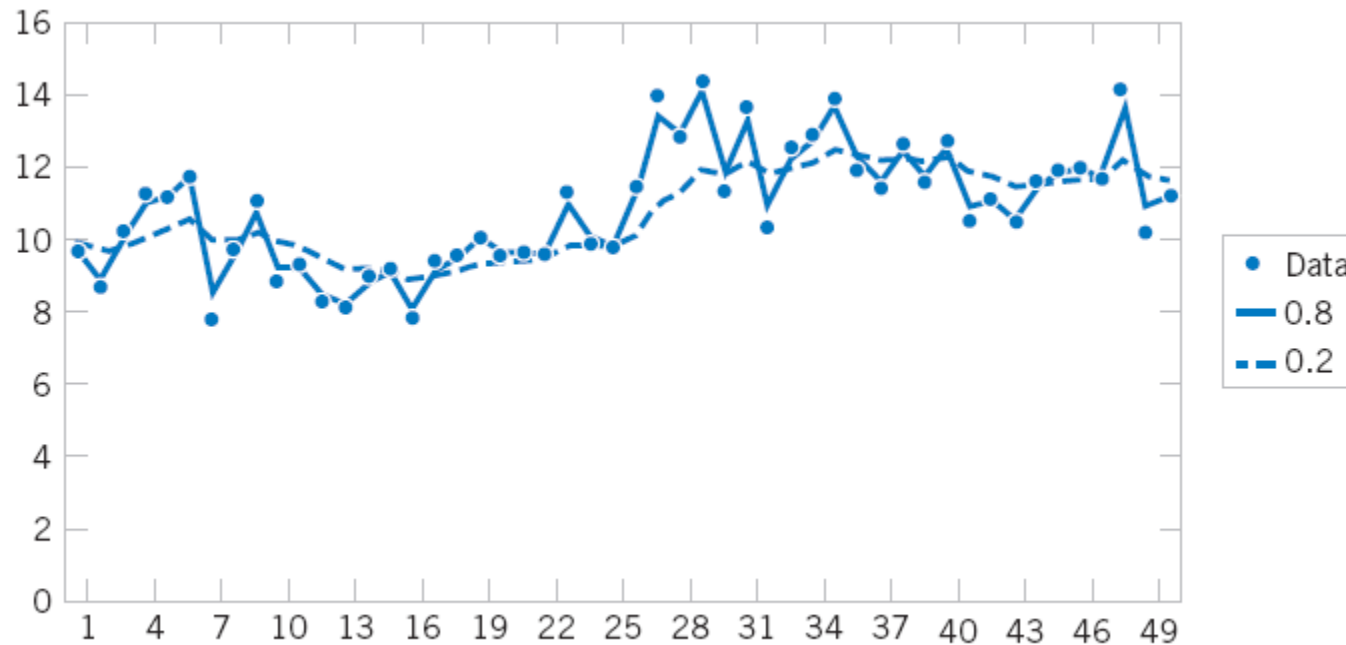
For statistical process control, rather than use a fixed window size it is useful to place the greatest weight on the most recent observation or subgroup average, and then gradually decrease the weight on older observations. One average of this type can be constructed by a multiplicative decrease in the weights. Let $\lambda \leq 1$ denote a constant and μ_0 denote the process target or historical mean. Suppose that samples of size $n \geq 1$ are collected and $\bar{x}_t$ is the average of the sample at time t . The **exponentially weighted moving average (EWMA)** is

$$\begin{aligned} z_t &= \lambda \bar{x}_t + \lambda(1 - \lambda) \bar{x}_{t-1} + \lambda(1 - \lambda)^2 \bar{x}_{t-2} + \cdots + \lambda(1 - \lambda)^{t-1} \bar{x}_1 + (1 - \lambda)^t \mu_0 \\ &= \sum_{k=0}^t \lambda(1 - \lambda)^k \bar{x}_{t-k} + (1 - \lambda)^t \mu_0 \end{aligned}$$

Each older observation has its weight decreased by the factor $(1 - \lambda)$. The weight on the starting value μ_0 is selected so that the weights sum to one. Here z_t is also sometimes called a geometric average.

Time-Weighted Charts

Exponential Weighted Moving Average Control Chart



EWMA with $\lambda=0.8$ and $\lambda=0.2$ show a compromise between a smooth curve and a response to a shift

Time-Weighted Charts

Exponential Weighted Moving Average Control Chart

EWMA Control Chart

$$LCL = \mu_0 - 3 \frac{\sigma}{\sqrt{n}} \sqrt{\frac{\lambda}{2 - \lambda} [1 - (1 - \lambda)^{2t}]}$$

$$CL = \mu_0$$

$$UCL = \mu_0 + 3 \frac{\sigma}{\sqrt{n}} \sqrt{\frac{\lambda}{2 - \lambda} [1 - (1 - \lambda)^{2t}]}$$

Time-Weighted Charts

Chemical Process Concentration EWMA

Consider the concentration data. Construct an EWMA control chart with $\lambda = 0.2$ with $n = 1$. It was determined that $\bar{x} = 99.1$ and $\overline{mr} = 2.59$. Therefore, $\hat{\mu}_0 = 99.1$ and $\hat{\sigma} = 2.59/1.128 = 2.30$. The control limits for z_1 are

$$LCL =$$

$$LCL =$$

The first few values of z_t along with the corresponding control limits are

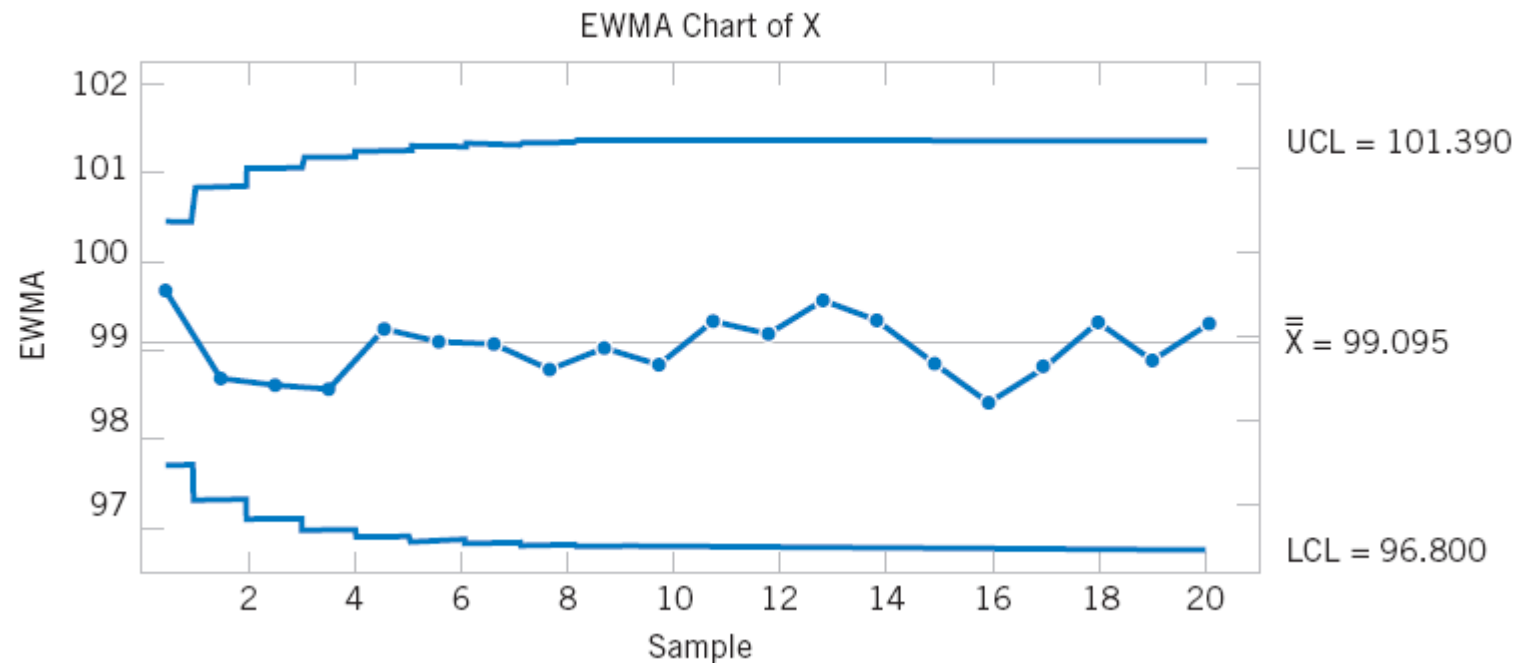
t	1	2	3	4	5
x_t	102.0	94.8	98.3	98.4	102.0
z_t	99.68	98.70	98.62	98.58	99.26
LCL	97.72	97.33	97.12	97.00	96.93
UCL	100.48	100.87	101.08	101.20	101.27

The chart generated by Minitab is shown in Figure.

Notice that the control limits widen as time increases but quickly stabilize. Each point is within its set of corresponding control limits so there are no signals from the chart.

Time-Weighted Charts

Exponential Weighted Moving Average Control Chart



EWMA control chart for the Chemical Process Concentration Data from Minitab.